

BIAI Project Card

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1. Subject of the project.

Image Classification of Vehicle Brands using Deep Learning.

2. Overview.

The main objective of this project is to develop, train, and evaluate a robust Deep Learning model capable of accurately classifying images into specific, visually similar categories of different car brands. The project addresses the complex challenge of fine-grained image classification that require deep feature extraction. Developing this solution will allow for the automation of vehicle recognition and categorization of brand proving the effectiveness of advanced neural networks in analyzing visual data.

3. Method of implementation.

The project will be implemented using the Python programming language. The core methodology relies on Transfer Learning using a Convolutional Neural Network architecture to extract deep visual features. To achieve high accuracy on visually similar objects, advanced metric learning techniques, will be applied to optimize the distance between image features in the latent space. The training methodology will also include Data Augmentation to prevent overfitting, Early Stopping mechanisms, and model explainability evaluation using Grad-CAM to visualize the neural network's decision-making process.

Data sources:

Publicly available image datasets from Kaggle

Programming Language:

Python 3.x

Core Frameworks:

TensorFlow, Keras, and PyTorch

Libraries & Solutions:

timm, TorchMetrics, pytorch-grad-cam, PIL, torchvision, numpy, OpenCV

Environment:

Google Colab, Jupyter Notebook, PC with Nvidia GPU for local training